5G Network Slice - A URLLC Resource Allocation Perspective

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***Abstract*—The fifth generation wireless network, known as 5G, is the first network that was considered to meet some requirements to help IoT become a reality. 5G is also envisioned as a network that will support a diverse set of performance and service requirements and help create a wide range of vertical markets. One of the main facilitators of 5G that will provide such flexibility is slicing a single physical network into multiple isolated logical networks, each with different requirements. Different technologies have been studied to make it possible to create this end-to-end network on demand, such as Software Defined Networking (SDN), Network Function Virtualization (NFV) and Cloud Computing. However, allocating enough resources to guarantee the desired QoS for specific services even when this virtual network fails remains forgotten by the literature. In this paper, we identify challenges and open issues related to this subject. Moreover, we discuss relevant proposals and point some possible directions to overcome this problem.**

***Index Terms*—5G, URLLC, Network Slicing, Resource Allocation**

I. Introduction

There is a growing number of devices with computing power, communication, and sensorial capacity surrounding us, mainly those equipped with wireless communication technologies. Their vast proliferation of these has helped collect a higher amount of information on numerous scenarios. This paradigm known as the Internet of Things (IoT) is not new to the academy nor the industry, although the network we have today was not thought to support this type of scenario. The fifth generation wireless network (5G) is the first thought to meet some requirements that will help IoT become a reality.

5G will not only help IoT, as it has already created a range of opportunities for applications and services. Considered a network that will support a diverse set of requirements concerning performance and service, 5G will help create a wide variety of vertical markets.

It is likely that in the future, 5G will be a union of evolutionary and service-oriented views. The first view focuses on scaling up and improving the efficiency of mobile networks by increasing traffic volume, the number of devices supported, throughput, etc., while the second one aims at offering a variety of different services with different requirements, that will encompass not only human type communications but also machine type communications [1].

5G desired capabilities are based on these two perspectives, and they are challenging because today's network is based on a one-size-fits-all idea, while 5G should be flexible, programmable, and use open interfaces. That will create a multi-tenant ecosystem that will, therefore, promote new business markets [2].

5G prerequisites must satisfy both traditional mobile communications and new vertical market segments, such as automatic driving, unmanned aerial vehicles (UAV), telemedicine, massive Internet of Things (mIoT), to mention a few. This variety of applications require particular networks with differentiated quality of service (QoS), service level agreements (SLAs), and key performance indicators (KPIs). This service-oriented vision leads to a notion of slicing the network on a per-service basis.

Network Slicing (NS) can be simplified to the idea of splitting a single physical network into multiple logical self-contained networks. This technology enables the operator to create, customize, and provide optimized solutions for multiple market scenarios with different specifications.

The idea of network slicing can also be compared to infrastructure-as-a-service (IaaS) in cloud computing, where computing, storage, and networking resources are shared among different tenants. Some works also compare it to the Network-as-a-service (NaaS) paradigm, and others consider it a new cloud paradigm called Slice-as-a-Service (SlaaS) [3].

In section II, we discuss the basic concepts necessary to understand this paper fully. We present 5G network requirements and its mains classes in Subsection II-A. Moreover, the definition of 5G Network Slice is shown in Subsection II-B. We finish this section by talking about the cloudification process of the network and how it is related to NS in Subsection II-C.

Section III discusses Ultra-Reliable and Low Latency Communications (URLLC) and its requirements. We also talk about the challenges of allocating resources for this type of slice and point out some directions to solutions to solve these issues at Subsection III-A.

Lastly, in Section IV, we present our final considerations regarding the resource allocation for Ultra-Reliable and Low Latency Communications slices and present some future work perspectives.

II. Background

As we stated in Section I, today's network was not thought to support an IoT scenario. Besides that, the diversity of emerging applications and their requirements demand a new network. With these challenges, the fifth generation is expected to have significantly faster speeds (between 1 and 10 Gbps), support numerous devices, reduce energy consumption and latency.

A. Fifth generation - 5G

5G represents the future generation of mobile telecommunication but, as stated before, it is not an incremental improvement of predecessors. Fifth-generation is meant to be an innovative network with unprecedented data rates, latency, connectivity, reliability, and energy efficiency [4]. There are eight major requirements for 5G [5], [6]:

- From 1 to 10 Gbps - It is an increase of almost ten times compared to the traditional LTE network's data rate.
- 1 ms round trip latency - A ten times decrease compared to 4G.
- High bandwidth within the same region- Allows numerous connected devices with higher bandwidths for longer durations in a particular area.
- An enormous number of connected devices - As stated, 5G networks need to provide connectivity to billions of devices.
- 99.999% availability - With 5G, the network will essentially be available all the time.
- Nearly 100% coverage for 'anytime anywhere' - IoT aims to access anything, anytime, anywhere, and for that, 5G wireless networks need to guarantee full coverage regardless of users' locations.
- Decrease around 90% of energy usage - Besides considering the development of green technology, energy usage will be even more crucial with high data rates and massive connectivity of 5G wireless.
- Long battery life - Reduce power consumption by devices is crucial in emerging 5G networks.

The fourth-generation was based on a one-size-fits-all idea, suitable for everyone or every purpose. However, to provide optimized support for the various categories of use cases/scenarios, 5G must be flexible and adaptable. The International Telecommunication Union(ITU) and Fifth Generation Public Private Partnership (5G-PPP) have identified the following requirements for different cases in three main categories [7]. Figure 1 and Table I present performance requirements for these categories.

- Enhanced mobile broadband (eMBB) - It encompasses what we have today but considering a superior performance and user experience. This category covers a variety of cases, including wide-area coverage and access points. For broad-range coverage, continuous coverage and medium or high mobility and an improved data rate are desired. The access points must provide support for high user density areas and a high volume of data traffic, considering that mobility is low. Also, the user data rate is higher than in comparison with wide-ranging coverage.
- Ultra-Reliable and Low Latency Communications (URLLC) - This use case has stringent requirements for features like flow, latency, and availability. This category encompasses intelligent transportation systems, remote medical surgery, public safety and emergency support, wireless control in industrial manufacturing, distribution automation in a smart grid, among others.
- Massive Machine Type Communications (mMTC) - This last category comprises scenarios where there is a massive number of connected devices that typically send a relatively low volume of non-delay sensitive data. In this case, the devices should be inexpensive and have long battery life.

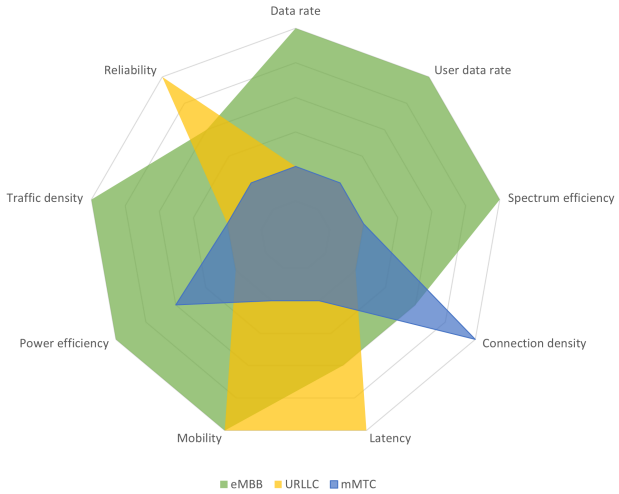


Fig. 1. The importance of key capabilities in different usage scenarios [1]

B. 5G Network Slicing

It is important to emphasize that these requirements are not only remarkably challenging but diverse. That is why a one-size-fits-all network will not be able to satisfy all these requirements.

One way to address this challenge of supporting the wide range of use cases is to use the network as a service paradigm, or NaaS. That is, logical (virtual) networks are created on the same physical infrastructure and are optimized to work in a specific use case. For example, eMBB applications require a large bandwidth to offer services with high data rates, such as streaming high resolution videos. Applications URLLC (eg, autonomous driving) need reliable, secure and low latency communication. In this way, a virtual network can be created for each of these categories and optimized to meet the requirements of your applications.

There seems to be a broad consensus on the need for native flexible and programmable networks that support softwarization and network slicing as a means of realizing widely different services in 5G. With SDN and NFV technologies all

KPI	Use cases	Values
Maximum data rate	eMBB	DLL: 20 Gbps, Uplink (UL): 10 Gbps
Maximum Spectral Efficiency	eMBB	DLL: 30 bps/Hz, UL: 15 bps/Hz
Latency of user plane	eMBB, URLLC	4 ms for eMBB and 1 ms for URLLC
Control Plane Latency	eMBB, URLLC	20 ms for eMBB and URLLC
Density of connection	mMTC	1,000,000 devices $/km^2$
Energy Efficiency	mMTC	Able to withstand long sleep periods (sleep mode) to allow low power consumption when there is no data
Reliability	URLLC	Succession of $1 - 10^{-5}$ to transmit a 32-byte data unit in a Layer 2 protocol in 1 ms with the quality of the channel at the edge of the cover
Mobility	eMBB	Up to 500km/h
Mobility interruption time	eMBB, URLLC	0 ms

TABLE I
KEY PERFORMANCE INDICATORS (KPI) FOR THE 5G USE CASES [8]

parts of the network architecture can be virtualized to form end-to-end virtual networks [1].

Various architectures were proposed and studied in the literature [1], [9], [10] and even though there is not a standard one, these solutions have similarities that can be mapped, at least at a high level, in three main layers and an entity as described below [1]:

- Infrastructure/resource layer. This layer provides a reserve (pool) of resources (computational, storage, network and radio) that can be virtualized and allocated to slices.
- Network function layer has a set of virtual and real network functions that run on the resources provided by the infrastructure layer.
- Service/application/business layer. This layer provides high-level descriptions of end-to-end network services and interfaces for creating new services.
- Management and orchestration entity. This entity is responsible for the implementation of slices using the chain of network functions and infrastructure resources. It also controls the entire life cycle of these slices.

Network slicing can improve resource efficiency and make it feasible to decouple network infrastructure and network services. That means that companies, e.g. mobile network operators (MNOs), do not need to run and maintain both infrastructure and services by themselves. It is possible to lease network slices to multiple tenants, e.g. service providers (SPs), that can create services and deliver them to the end customers without possessing their own network infrastructure, as illustrated in Figure 2. This business model, where network slices are provided as a public cloud service, is known as slice-as-a-service (SlaaS) [3]. Again, it is clear that network slicing and cloud computing have some similarities.

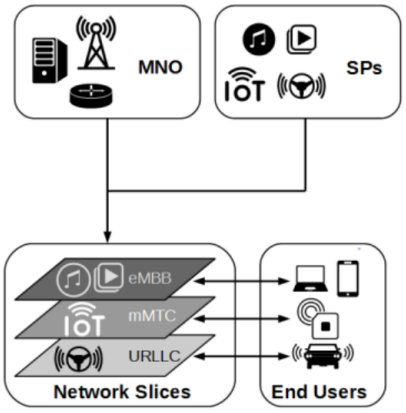


Fig. 2. Multi-tenancy sliced network [11]

Since there are multiple definitions for network slicing in the literature, we will adopt the following definition throughout our work for the sake of simplicity. A Network Slice is an end-to-end logical network composed of different resources from a physical infrastructure that can belong to different companies and its resources can be shared among different tenants. Each network slice is an instance of an end-to-end logical network that is independent and isolated from each other.A slice can be thought of as a set of subsets, which means we have allocate enough resources on all parts of the network to meet the desired requirements for that specific slice. This proves, once more, the similarities between network slicing and cloud computing.

C. Network cloudfication

As defined by the National Institute of Standards and Technology (NIST), cloud computing is a model that enables ubiquitous on-demand access to a shared pool of configurable computing resources (e.g., networks, storage, applications, and services). These resources can be rapidly deployed and released with minimal management effort or service provider interaction. The cloud offers computing, software, and application resources as a utility the same way a cloud-native network does.

Cloudification can be defined as the process of bringing virtual elements to the cloud. Network cloudification can be achieved by using mainly two approaches [12]:

- Cloud-based: The most straightforward cloudification process. This approach aims to move the conventional network architecture to the cloud, recognized as the Network-as-a-Service model. A single instance is deployed in each service, and the orchestrator ensures it is performing properly. However, since the standard solution for problems is to restart or create a new instance and each service has only one instance, this solution will lead to an SLA violation.
- Cloud-native: This approach aims to develop and run the applications and services while using all the advantages

of cloud computing. Cloud-native services are expected to incorporate the cloud-like model (e.g., scale up and down resources while guaranteeing the desired availability, base on demand).

Despite being a more straightforward, more comfortable designing and implementing approach, cloud-based comes up with several limitations, such as availability, scalability, and flexibility, which are essential for 5G network slicing, especially for URLLC slices. On the other hand, the cloud-native model helps companies and users implement and develop new ideas faster, which reduces their time to market the same way cloud computing does.

III. URLLC Resource Allocation

URLLC is an essential new feature brought by 5G, with the potential to support a vast set of applications that rely on mission-critical networks. Ensuring low-latency and ultra-reliable communication for future wireless networks is of capital importance. However, Network Slice models and implementations in the literature do not consider this use case or consider important points such as reliability and availability.

Reliability can be defined as the capability of transmitting a given amount of traffic within a predetermined time duration with high success probability [13]. Besides, according to [14], reliability is the continuity of service; in other words, the probability that a system or service remains operable for a specified interval of time. Furthermore, availability is the probability that a system or service will be operable when needed [13].

Both are important to achieve resilience, which is the ability of the network to provide and maintain an acceptable level of service in the face of various faults and challenges to standard operations [13]. Therefore, URLLC slices must be designed to be resilient, which means that enough resources must be allocated to create a slice that provides service in the face of challenges, such as a network link failure, collisions between users due to channel access coordination, virtual network function failure, overloaded edge cloud, among others.

In [9] the authors classify different models of resource allocation algorithms based on their objectives: i) economic/game model; ii) prediction model; iii) robust/failure and recover model. As pointed out, most of the works fall into the first two categories, mainly because URLLC slices require additional backup resources to be used in case of failure and because most customers do not need or are unwilling to pay extra for high levels of availability.

A. Challenges and Directions

In this Section, we will discuss some challenges that affect the instantiation of URLLC slices. We will also present possible directions to overcome these issues.

1) Cost: This is the primary factor affecting the amount of redundancy and fault-tolerant mechanisms used in a network slice. This limitation affects network resilience by limiting the number of resources that an operator can afford to add to the network infrastructure [15]. If we consider, for example, dedicated protection, we need to double the number of resources for every slice, which might lead to overpricing or a scarcity of resources.

Some works such as [16] use techniques as bandwidth squeezing and survivable multi-path provisioning to overcome this problem. However, it is important to mention that this still might lead to an SLA violation.

One possible solution to this matter is to support resilience classes since not all URLLC slices have the same requirements. Therefore, a categorization of this class can be used to provide different levels of availability and fault protection. For example, using dedicated protection only for the most critical applications and shared protection for less demanding ones.

2) End-to-end infrastructure: An end-to-end network will typically traverse several networks, which means they can traverse an infrastructure that belongs to multiple owners and operators. One more time, end-to-end availability might be compromised.

To solve this issue, it would be necessary to share information among different providers, possibly competing ones. Fortunately, some works considered this and presented solutions for multi-domain slice orchestration like [17]. Some projects such as [18] consider federations to form end-to-end multi-domain slices where resources are shared and coordinated by the federation.

A multi-level orchestration might also be the answer to this problem. For example, a master orchestrator might use the information of lower-level orchestrators to check the availability of resources in multiple infrastructures and decide which is the best option.

Another possible solution is to pay as much as possible for the service, which leads us back to our first challenge.

3) Shared Risk Link Groups: In traffic engineering, a Shared Risk Link Group (SRLG) is a set of links sharing a common resource, which affects all links in the set if the common resource fails. Since these resources share the same risk of failure, if multiple slices are using them, we need to migrate or allocate multiple resources at the same time to recover the services.

One possible solution to this case is to avoid using common resources for URLLC slices since they have restricted requirements and are dedicated to critical applications. For example, if a resource is shared between a URLLC and mMTC service, the orchestrator will prioritize the critical mission service over the machine type one.

4) Risk Based Design: Another issue that should be taking into account is the likelihood of failures. In other words, observe failure and repair rate of resources, check if they are geographically correlated due to weather, workforce capabilities, exposure to natural disasters, among others.

This type of study might be necessary during the allocation process of critical mission slices. Providers might use this information to avoid using resources that are already known for their instability and assign them to less demanding slices such as mMTC.

This information can also be used to analyze how these failures impact the QoS and create different risk classes since URLLC services may have different requirements.

5) Energy efficiency: Since we expect to reduce the energy use on 5G, we cannot afford to waste resources in order to guarantee resilience and availability for critical mission applications.

Once more the idea to share this unused pool of resources seems tempting, especially if we consider that we can use them for less demanding classes.

6) Overprovisioning/underprovisioning: Lastly, the most important challenge is the dilemma between overprovisioning and underprovisioning the necessary resources to guarantee the desired requirements for a URLLC slice. This challenge is intimately related to all the other ones.

Overprovisioning is the simplest way to guarantee that all the requirements will be met. However, as we already stated, this will lead to a scarcity of resources, high costs for the tenants, and low energy efficiency. On the other hand, underprovisioning will lead to more significant risks of failure that will lead to SLA violations, which is already what most of the models do when ignoring critical mission services requirements.

The following options might address this issue: i) use of class differentiation inside the URLLC class as mentioned before; ii) path squeezing and multi-path provisioning as shown in [16]; iii) shared protection to reduce resource scarcity as shown in [19] and can be used in combination to class differentiation; iv) heuristic algorithms might optimize resource utilization as shown in [20]; and v) machine learning to scale resources up and down in a fast manner as shown [21], so there is no need to backup resources in standby.

IV. Conclusion

In this paper, we presented 5G network requirements, their class differentiation and talked about cloudification of the network. Furthermore, we talk about Ultra-Reliable and Low Latency Communications (URLLC) and its requirements and explain why most works in the literature will not be suitable for this class type. Finally, we present some challenges imposed by URLLC networks and present some directions on how to overcome them.

We understand that classic solutions, such as dedicated protection, will lead to excessive prices for these services or that providers will avoid offering these services due to a lack of resources to meet the demand. Moreover, we believe that it is possible to design solutions that respond proactively and are affordable for tenants and operators, and we plan to pursue this subject in future works.

URLLC slices or critical mission slices are an area rich in open research problems that need to be studied to achieve 5G fully potential.

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